Department of Computer Science and Engineering
Academic Year 2021 - 2022 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. Computer Science and Engineering
Course Code & Title: CS8451 & Design and Analysis of Algorithms
Name of the Faculty member (s): Ms.P.Jothi Thilaga, AP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit IV / Linear Programming - Simplex Method
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO8
- **Activity Chosen:** Flash Cards
- **Justification:**

  Closest-Pair and Convex Hull problem is one of the important topics, repeatedly asked in university questions. Minute paper is a short writing activity within the classroom in which students generate answers in response to questions asked by the teacher regarding the concept of closest pair and convex hull algorithms taught in class. Its major advantage is that it provides rapid feedback on whether the instructor's main idea and what the students perceived as the main idea are the same.
- **Time Allotted for the Activity:** 5 Minutes
- **Details of the Implementation:**

  A flashcard is a method in which the questions and corresponding answers are in separate cards. Question and answer cards are prepared on sorting techniques as shown in Figure 1, taught in the previous day's class. Initially, Flashcards containing questions and answers were distributed to students in the class. One student from the class displayed the question card, the other students in the class identified the cards relevant to the questions within a minute. Then the students explained the answer in front of the classroom for the next 2 minutes with their answer cards, relevant to the corresponding question as shown in Figure 2. This activity helped the students to revise the simplex method to solve linear programming problem in unit IV.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO10	PSO1	PSO2	PSO3
CO1	3	2	1	1	2	2	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO4**
	3	2	1	1
Justification for correlation	Linear Programming problem is a basic concept needed for many applications.	These concepts are used in analysis of solving engineering problems	To analysis complex problems and represent the solution using this concepts	To provide the solution for complex problem using simplex method of linear programming.
	PO10	**PSO1**	**PSO2**	**PSO3**
	2	2	1	1
	Able to represent the solutions for the linear programming problem using simplex method	Able to use these concepts in the applications of Information Technology	Able to use simplex method in specific areas for developing reliable IT Solutions	Simplex method is helpful in solving real world problems in Research.

- **Images / Screenshot of the practice:**

Figure 1: Flash Cards Activity

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - Use of minute papers in the classroom actually increases attentiveness in the class and also improves the students' skill of writing and critical analysis of the topics.
    - This activity reflects the understanding and involvement of the students in the particular topic.
  - ***Benefit of the practice:***
    - Students were keenly participated in this activity.
    - From this activity, each student can refresh the closest pair and convex hull concept and understood the topic clearly while the information is discussed and recalled.
  - ***Challenges faced in implementation:***
    - Some students are hesitated and lack to share the concepts that they learnt in the class.
    - Slow learners were felt difficult to write the key points in the topic.

References:

- https://www.rochester.edu/college/cetl/faculty/one-minute-paper.html
- https://www.unl.edu/gradstudies/current/teaching/minute
- https://oncourseworkshop.com/self-awareness/one-minute-paper/

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